What does Empirical Evidence demonstrate about Algorithmic Choice?

An Empirical Demonstration of the Limits of Algorithmic Selection

Nuño Valenzuela Alonso

Philosopher and Researcher in Algorithmic Decision-Making and Formal Methods
ORCID 0009-0001-6897-3912

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Abstract

What Does Empirical Evidence Demonstrate About Algorithmic Choice? This paper provides a controlled empirical examination of the distinction between an algorithmic choice, the objective quality of that choice, and the evidence available to justify a comparison between alternatives. Using finite MIN/MAX evaluation structures and archived UCT/PUCT search outcomes, we present explicit counterexamples in which the selected action is not the objectively optimal one. These counterexamples do not establish population-level error rates; rather, each constitutes a witness against any universal optimality claim whose domain includes the corresponding case.

More importantly, we demonstrate that complete information is not necessary to establish that one alternative is strictly better than another. In a finite evaluation instance, only five observations suffice to certify strict superiority across every completion compatible with the available evidence. Thus, incomplete information need not imply comparative indeterminacy: what matters is not whether the information is complete, but whether the available evidence is sufficient to determine the comparison. Conversely, removing one critical observation destroys this guarantee, allowing two completions compatible with the remaining evidence to produce opposite rankings.

Finally, an archived fixed-seed trajectory shows that increasing computational budget need not monotonically improve the quality of the selected action along a particular run. Together, these results empirically separate algorithmic selection, objective optimality, and evidential sufficiency. An observed algorithmic choice does not by itself establish that the selected alternative is best; conversely, incomplete evidence can nevertheless be sufficient to prove that one alternative is better than another.

The article is self-contained with respect to its numerical demonstrations. The archived UCT/PUCT selections are empirical witnesses, not independently re-executed trajectories; full execution-level reproduction requires the original implementations and configurations.

Keywords: algorithmic selection; computational evidence; UCT; PUCT; minimax; partial evidence; counterexample; verifiability; formal logic.

1 The question and the claim

An algorithm can return an answer without having established that no better answer exists. This distinction is familiar in logic: a recorded choice is a fact about the execution, whereas an optimality claim requires a comparison under a declared objective. The companion studies distinguish observed selection from demonstrated preference and characterize exact comparisons

based on partial information [2, 3]. The present article asks a more focused empirical question: *which claims can be checked against a preserved execution record and a fully disclosed numerical reference?*

We report two empirically grounded, independently inspectable results. In the first, preserved algorithm runs selected particular actions; their full synthetic decision trees are summarized numerically, allowing an exact comparison with the selected actions. In the second, the original decision is set aside: five values of terminal nodes support a universal comparison over the incomplete trees compatible with those observations. These are different claims. A historically recorded action must not be confused with a proved comparison, and a certificate constructed after a run must not be attributed to the algorithm during that run.

Organization. We first define the reference objective and the limits of the preserved observations. We then display the numerical tree evaluations and four historical selections, followed by a positive and a negative certificate under partial information. The instance names *aligned*, *deceptive*, and *optimal-last* are descriptive names for synthetic trees, not new algorithms or theoretical categories. Archival package labels are used only as provenance identifiers, never as evidence of scientific validity.

2 Empirical scope and validation criteria

2.1 What the experimental records are — and are not

This is a *finite-witness study*, not a population study. The four algorithmic selections are previously archived outcomes of individual runs under fixed input trees, search configurations and finite computational budgets. Their common pseudorandom seed, 0, makes them instances of a controlled configuration; it does not create independent statistical replicates. Accordingly, the article estimates neither the prevalence of suboptimal decisions nor an average failure probability. Each recorded outcome is treated as a concrete *empirical witness*: a fact about one specified execution that can be compared with an independently defined reference value.

The three trees are *synthetic audit artifacts*: small, fully specified numerical examples designed to expose the logical difference between a returned decision and its objectively evaluated value. A synthetic case is not evidence of performance on deployed systems or natural-world data. Its scientific role is to make the full reference computation inspectable.

2.2 Three distinguishable validation claims

A careful reviewer should not treat program execution, numerical verification, and logical inference as interchangeable. Table 1 records what constitutes evidence for each.

Source and provenance. The four selected actions and their finite-budget settings are inherited from preserved UCT/PUCT execution records examined in the earlier experimental audit. The reference trees, their terminal payoffs, and the retrospective five-observation case derive from the corresponding archived finite-tree datasets. This manuscript performs no new search runs and does not combine dependent records into additional replicates. It reports numerical values as transcribed evidence and exposes the arithmetic needed to check the conclusions. The distinction is important: the paper supplies an inspectable calculation, not the implementation or environment needed to rerun historical search trajectories.

These are operational checks adopted for this study, not a claim of accreditation under a software testing standard. *Traceability* means the reader can follow each inference from a stated observed input or archival selection through an explicit calculation to a conclusion of no greater

Table 1: Validation layers, acceptance checks and limits of each claim.

Layer	Evidence and acceptance check	What it does not establish
Archived execution	A preserved record identifies the method, input tree, seed, finite budget and selected action.	Independent re-execution or representative error frequency.
Exact reference evaluation	Terminal values are combined by the declared MAX–MIN rule. Acceptance: a positive gap $\Delta(b)$ for the recorded choice.	A mechanism causing the choice or general failure of a search method.
Partial-information certificate	Observations and admissible intervals yield exact bounds. Acceptance: $L_E(a_5) > U_E(a_0)$, with a reversal after deletion.	A certificate possessed by the search procedure at decision time.

strength. The sections below make the complete numerical checks visible without requiring scripts.

3 Terms and elementary evaluation rules

3.1 Decision tree, action and terminal value

A *decision tree* is a finite arrangement of possible actions and subsequent outcomes. The decision maker, called MAX, seeks a larger numerical result; its opponent or adverse continuation, MIN, seeks a smaller one. A *terminal value* is the numerical payoff assigned to an outcome at the end of a branch. There are six legal initial actions $a_0, \dots, a_5$ in each example, not six competing algorithms.

Each action has three possible adverse replies, labelled r_0, r_1, r_2. Each reply ends in a MAX choice between two terminal values x_{ir0}, x_{ir1} (with i denoting the initial action and r the reply). The completed-tree evaluation is

$$M_{ir} = \max(x_{ir0}, x_{ir1}), \qquad V^*(a_i) = \min_{r\in\{0,1,2\}} M_{ir}. \tag{1}$$

The star in V^* denotes evaluation of the *fully specified* tree, not evidence that the algorithm knew the answer. The first operation takes a maximum within each reply; the second takes the minimum over the three replies. Thus a reader can reproduce each action value using only maxima and minima.

3.2 UCT, PUCT, budget and observed selection

UCT (Upper Confidence bounds applied to Trees) is a sampling-based tree-search method that balances exploration of alternatives with exploitation of estimated performance. *PUCT* (predictor-guided UCT) is a related selection rule that incorporates prior action weights. Their sampling-based estimates are not automatically exact MIN/MAX values. The *budget* B in the archived examples is the permitted number of root-to-terminal simulations (rollouts). The *seed* fixes the pseudorandom sequence, and the *action order* fixes how alternatives are enumerated. A *selected action* b is the action returned in the archived execution. These terms describe how the recorded choice arose; they do not themselves prove that b was optimal.

A comparison with the completed tree uses the *optimality gap*

$$\Delta(b) = \max_{a\in\{a_0,\dots,a_5\}} V^*(a) - V^*(b). \tag{2}$$

If $\Delta(b) > 0$, a better legal action exists in this specified tree. The size of the gap measures the difference in terminal value, not an error probability.

3.3 Evidence, bounds, certificate and completion

A *partial observation* tells us the values of some leaves but not others. An unobserved leaf is allowed to take any value in its stated admissible interval, here $[-1, 1]$. A *completion* assigns one admissible value to every unknown leaf, without changing the observed leaves. The set of all these assignments is $\mathrm{Compat}(E)$, where E means the evidence retained. The variables are independently assignable within their bounds; this is *not* a claim of statistical independence. If a variable appears more than once, its value must be consistent in every occurrence.

For an initial action, the smallest and largest possible values over all completions are

$$L_E(a) = \min_{\gamma \in \mathrm{Compat}(E)} V_\gamma(a), \qquad U_E(a) = \max_{\gamma \in \mathrm{Compat}(E)} V_\gamma(a). \tag{3}$$

An *evidence certificate* is a finite, inspectable set of observations and valid calculations that justifies a stated conclusion. In particular,

$$L_E(b) > U_E(a) \quad \Longrightarrow \quad \forall \gamma \in \mathrm{Compat}(E),\ V_\gamma(b) > V_\gamma(a). \tag{4}$$

The right-hand statement is stronger than saying that b happened to outperform a once: it holds for *every* completion allowed by the evidence.

4 First empirical result: selections compared with exact tree values

4.1 All six action values are visible in the paper

The following three tables present, for every action, the results of the inner MAX operations, M_{i0}, M_{i1}, M_{i2}. The final column applies the outer MIN rule in Eq. (1). They summarize the original terminal leaves without hiding the criterion by which their exact strategic values are obtained. For example, in the aligned tree, action a_0 has a first reply with terminal values $(0.900, 0.801)$; the maximum is 0.900. Its remaining two reply maxima are 1.000 and 1.000. Thus $V^*(a_0) = \min(0.900, 1.000, 1.000) = 0.900$. The full pairs of terminal payoffs for actions cited in the four witnesses appear in Table 5.

Table 2: Aligned synthetic tree: MAX value after each of three replies, followed by the action's MIN value.

Action	M_{i0}	M_{i1}	M_{i2}	$V^*(a_i)$
a_0	0.900	1.000	1.000	0.900
a_1	0.650	0.803	0.839	0.650
a_2	0.450	0.603	0.639	0.450
a_3	0.200	0.353	0.389	0.200
a_4	0.000	0.153	0.189	0.000
a_5	-0.250	-0.097	-0.061	-0.250

4.2 The archived choices and their numerical consequences

Table 6 combines two different types of evidence: the *selected action* is transcribed from the preserved execution record, while the *true action values* follow from the displayed tree calculations. All four runs use seed 0 and the canonical action order $a_0, a_1, \ldots, a_5$. The reported PUCT

Table 3: Deceptive synthetic tree, evaluated with the same MAX–MIN rule.

Action	M_{i0}	M_{i1}	M_{i2}	$V^*(a_i)$
a_0	0.900	1.000	1.000	0.900
a_1	0.700	0.853	0.889	0.700
a_2	0.400	0.553	0.589	0.400
a_3	0.100	0.253	0.289	0.100
a_4	-0.200	-0.047	-0.011	-0.200
a_5	-0.400	-0.247	-0.211	-0.400

Table 4: Optimal-last synthetic tree: the optimal legal action is last in the canonical ordering.

Action	M_{i0}	M_{i1}	M_{i2}	$V^*(a_i)$
a_0	-0.300	-0.147	-0.111	-0.300
a_1	-0.100	0.053	0.089	-0.100
a_2	0.100	0.253	0.289	0.100
a_3	0.350	0.503	0.539	0.350
a_4	0.600	0.753	0.789	0.600
a_5	0.950	1.000	1.000	0.950

Table 5: Terminal values supporting the specific action comparisons (two terminal leaves for each adverse reply).

Instance	Action	Reply r_0	Reply r_1	Reply r_2
Aligned	a_0	$(0.900, 0.801)$	$(1.000, 0.954)$	$(1.000, 0.990)$
Aligned	a_1	$(0.650, 0.551)$	$(0.803, 0.704)$	$(0.839, 0.740)$
Deceptive	a_0	$(0.900, 0.801)$	$(1.000, 0.954)$	$(1.000, 0.990)$
Deceptive	a_2	$(0.400, 0.301)$	$(0.553, 0.454)$	$(0.589, 0.490)$
Optimal-last	a_1	$(-0.100, -0.199)$	$(0.053, -0.046)$	$(0.089, -0.010)$
Optimal-last	a_5	$(0.950, 0.851)$	$(1.000, 1.000)$	$(1.000, 1.000)$

variant for the deceptive tree uses uniform prior weights. The runs used the indicated small rollout budgets, not exhaustive strategic search.

Table 6: Four preserved decisions and independently checkable optimality gaps.

Instance	Search method	B	Selected b	Better action a	$\Delta(b)$
Aligned	UCT	2	a_1 (0.65)	a_0 (0.90)	0.25
Aligned	PUCT	2	a_1 (0.65)	a_0 (0.90)	0.25
Optimal-last	UCT	2	a_1 (-0.10)	a_5 (0.95)	1.05
Deceptive	PUCT, uniform	1	a_2 (0.40)	a_0 (0.90)	0.50

For instance, in the optimal-last tree the returned a_1 has value -0.10, while the legal action a_5 has value 0.95. The reader can calculate both values from Table 4; therefore $\Delta(a_1) = 0.95 - (-0.10) = 1.05$. This is a demonstrable *error of that recorded selection relative to the stated minimax objective*. It is not evidence that the search algorithm falsely asserted optimality, nor does it show that the algorithm had examined and rejected a_5.

Logical consequence. If a universal claim says that a particular implementation returns a maximally valued action on *every* run in a specified domain, one qualifying counterexample is enough to refute it:

$$\exists I \in \mathcal{D} :\ \Delta_I(b_I) > 0 \quad \Longrightarrow \quad \neg(\forall I \in \mathcal{D} : \Delta_I(b_I) = 0). \tag{5}$$

This is a standard counterexample argument. It does *not* establish an error rate, a convergence rate, or universal inferiority of UCT or PUCT.

4.3 A falsifiable witness of non-optimal selection

The logical condition for a counterexample is not merely that two reported numbers differ. It is the conjunction of an archived selection b and a fully specified reference tree containing a legal alternative a with strictly greater value:

$$\text{Selected}_h(b) \ \wedge \ \exists a \in A(s) : V^*(a) > V^*(b). \tag{6}$$

Here the observed selections are evidence of past computational executions; the inequalities are verifiable by the numerical reference rule. The four calculations are $0.90 - 0.65 = 0.25$ (UCT, aligned), $0.90 - 0.65 = 0.25$ (PUCT, aligned), $0.95 - (-0.10) = 1.05$ (UCT, optimal-last) and $0.90 - 0.40 = 0.50$ (PUCT, deceptive). Each positive gap establishes failure to maximize the reference objective on that particular recorded run. The witnesses refute any universal correctness claim whose domain includes those precise runs. They do not discredit the general heuristic, establish a population error rate, or show the search procedure represented its result as provably optimal.

4.4 A finite non-monotone budget trajectory

A separate archived budget comparison studies the *same aligned tree*, canonical action order, and fixed seed while changing the permitted number of simulations. Table 7 shows three observations in that record; they are not independent seeds. At budget $B = 1$ the selected action had exact value 0.90; at $B = 2$ its value was 0.65; at $B = 8$ the selected value returned to 0.90. This is a concrete, finite non-monotone trajectory of the *quality of the selected action*, not a statement about an algorithm's asymptotic convergence. The larger 32-row preserved grid spans two trees, two search configurations and eight budgets $B \in \{1, 2, 4, 8, 16, 32, 64, 128\}$, with one fixed

Table 7: Budget-dependent action values in the same aligned synthetic instance (archived configurations, seed 0).

Rollout budget B	1	2	8
Selected action	a_0	a_1	a_0
Exact MIN/MAX value	0.90	0.65	0.90
Gap to optimal	0	0.25	0

seed and order. Earlier archival checks reproduced its recorded rows; they are not additional independent trials in this paper. The finite result above neither estimates the likelihood of deterioration nor asserts that allocating more computation is generally harmful. Informative PUCT priors in related runs may be oracle-derived and must not be mistaken for independent information acquired by the search.

5 Second empirical result: five observations determine a comparison

5.1 The complete question under partial information

Consider only a_0 and a_5 from the optimal-last tree. We now *hide* all their terminal values except the five observations in Table 8. Every hidden leaf may independently take any value in $[-1, 1]$. This construction is retrospective: the observations were selected from the completed synthetic tree for a subsequent certificate, not necessarily observed by UCT or PUCT while choosing an action.

Table 8: The five retained terminal observations forming evidence E^+.

Initial action	Adverse reply and terminal branch	Known terminal value
a_0	r_0, branch c_0	-0.300
a_0	r_0, branch c_1	-0.399
a_5	r_0, branch c_0	0.950
a_5	r_1, branch c_0	1.000
a_5	r_2, branch c_0	1.000

5.2 Calculation of the bounds, without computer code

For a_0, the first adverse reply is fully observed, so its MAX value is $\max(-0.300, -0.399) = -0.300$. Either of the other two replies may range over $[-1, 1]$. Since the overall action value is the MIN over three replies, its smallest possible value is -1, and its largest is -0.300:

$$V_\gamma(a_0) = \min(-0.300, M_{01}, M_{02}) \in [-1, -0.300]. \tag{7}$$

For a_5, each of the three replies has one observed terminal value. The first reply's MAX is $\max(0.950, u)$, where $u \in [-1, 1]$, hence it lies in $[0.950, 1]$. The other replies have a known terminal value 1, so their MAX values are exactly 1. Thus

$$V_\gamma(a_5) = \min(\max(0.950, u), 1, 1) \in [0.950, 1]. \tag{8}$$

Both endpoints are attainable under the declared independent interval assignments. Therefore the exact bounds are

$$L_{E^+}(a_5) = 0.950 > -0.300 = U_{E^+}(a_0). \tag{9}$$

No knowledge of the hidden terminal values was used in these inequalities. For *every* compatible completion, $V_\gamma(a_5) \geq 0.950 > -0.300 \geq V_\gamma(a_0)$. This is an explicit numerical certificate, not an appeal to software or a sample of possible worlds.

5.3 Remove one observation: the conclusion no longer follows

Delete the observed value 0.950 at (a_5, r_0, c_0), leaving the other four observations unchanged; call the reduced evidence E^-. Both leaves under a_5's first reply are now unknown, so its MAX can take any value in $[-1, 1]$. The other replies still have MAX value 1; hence

$$V_\gamma(a_5) \in [-1, 1], \qquad V_\gamma(a_0) \in [-1, -0.300]. \tag{10}$$

The exact lower bound -1 of a_5 is no longer greater than the upper bound -0.300 of a_0. More decisively, Table 9 constructs two complete assignments compatible with the *same four observations* that reverse the preference. To obtain the upper or lower values of an unobserved reply, assign its two leaves to the indicated endpoint; all other unknown leaves can be assigned admissibly. These two witnesses demonstrate *non-identifiability* from E^-: the reduced observations

Table 9: Two compatible completed trees after removal of the 0.950 observation.

Completion	$V_\gamma(a_5)$	$V_\gamma(a_0)$	Difference	Comparative outcome
γ_+: choose unobserved values to favor a_5	1.000	-1.000	2.000	$a_5 > a_0$
γ_-: choose unobserved values to disfavor a_5	-1.000	-0.300	-0.700	$a_5 < a_0$

alone do not decide the ordering. Removing any one of the five observations also eliminates the strict endpoint inequality, as documented by the original five deletion calculations. That supports *inclusion-minimality* of this particular five-value certificate, not a claim that no different four-value certificate exists anywhere in a different tree or evidential setting.

5.4 Five deletion checks: the scope of minimality

The five-observation certificate is *inclusion-minimal* for this evidence set: if any one of the five terminal observations is removed and all remaining leaves range independently over $[-1, 1]$, the strict endpoint test no longer passes. Table 10 provides the complete five-case calculation. A failed endpoint test alone need not imply a reversal in arbitrary models; the explicit completions in Table 9 provide that stronger non-identifiability witness for the specific deletion of 0.950. This statement is deliberately narrower than global minimality: it does not assert that five

Table 10: Each one-observation deletion eliminates strict separation of exact bounds.

Removed observed leaf	$L(a_5)$	$U(a_0)$	$L(a_5) > U(a_0)$?
$a_0, r_0, c_0 = -0.300$	0.950	1.000	No
$a_0, r_0, c_1 = -0.399$	0.950	1.000	No
$a_5, r_0, c_0 = 0.950$	-1.000	-0.300	No
$a_5, r_1, c_0 = 1.000$	-1.000	-0.300	No
$a_5, r_2, c_0 = 1.000$	-1.000	-0.300	No

observations are necessary for every possible comparison, every alternative certificate, or every decision tree.

6 What the remaining preserved results say

As a bounded negative check rather than an additional positive finding, the research archive also contains a larger matrix of 77,952 pairwise comparison *records*, associated with 15,904 identified runs. These are not 77,952 independent trials. Of the pairwise records, 76,160 involved overlapping statistical intervals derived from rollouts, and 1,792 involved positive comparisons of exact terminal values. The declared investigation did *not* find a positive, non-trivial strategic certificate constructed entirely from admissible deterministic partial bounds in that archived domain. This is a negative result, not a proof that such certificates can never exist. The retrospective five-observation certificate of the preceding section does not change that result, because its observations were recovered after the original decisions.

A separate examination of the 32-run grid did not supply the variation required to assess the predictive usefulness of the proposed CES indicator. Here *CES* denotes an internal comparative-evidence score; its label must not be mistaken for a widely established measure or for an externally validated predictor. Two proposed transferability studies likewise remained blocked by unavailable relation contracts and matched inputs. Neither is treated as a successful experiment.

7 Discussion for a reviewer in logic and algorithm assessment

What is observed? The archived execution record says that UCT or PUCT returned one of the six actions at a declared budget and seed. The record is empirical. The present paper transparently transcribes four such facts but does not claim that a reader can independently reconstruct the search code from a table of results.

What is calculated? For each displayed fully specified tree, the numbers $V^*(a_i)$ follow by the elementary MIN/MAX rule. Thus the comparisons in Table 6 can be independently checked from the printed payoffs. The exact numerical result is contingent on the tree and the declared objective; it is not a conclusion about every algorithmic decision.

What is established from incomplete evidence? The five retained terminal observations and the allowed interval $[-1, 1]$ imply a strict comparison in *every* compatible completion. The paper prints the observations and the full bound calculation. The two four-observation completions refute any claim that those same reduced observations settle the ranking.

What does this mean for responsible evaluation? Scientific, institutional, and European governance assessments benefit from separating an output, its trace, its retrospective ground truth, and the comparative conclusion actually warranted by the evidence. The results illustrate a discipline of traceability and proportionate claim-making. This separation is relevant to the objectives of traceability and testable claims in algorithmic assessment, including European discussions of responsible AI. It does not itself establish conformity with EU AI legislation, certification under any V&V standard, formal ethical correctness, real-world reliability, or approval of a deployed system.

7.1 Claim-to-evidence traceability

Table 11 identifies each conclusion, the numerical evidence on the page, and its appropriate inferential boundary. This is a reporting device for scientific review, not a claim of formal accreditation.

Table 11: Traceability from observation and calculation to warranted claim.

Claim	Checkable evidence	Boundary
The archived choices are suboptimal on the specified trees.	Selected actions and V^* values in Tables 6 and 5; four positive gaps.	Does not estimate error probability or establish a universal algorithmic failure.
Budget-dependent quality need not improve at every measured budget.	Table 7: $0.90 \to 0.65 \to 0.90$.	One fixed-seed finite trajectory, not a convergence law.
Five observations entail strict superiority.	Table 8 and Eq. (9).	Holds for the stated fixed tree and admissible completions; is not contemporaneous UCT/PUCT evidence.
Four observations do not settle the same pairwise comparison.	Two opposite compatible completions in Table 9.	Does not imply the opposite ranking is universally true.

8 Limits of reproducibility and conclusion

The paper is self-contained with respect to its *numerical demonstrations*: the printed definitions, tables, and arithmetic suffice to inspect the four gaps, the five-observation certificate and its specified deletion countermodels. No Python program or proprietary research software is necessary to check these mathematical statements. The preserved actions themselves are historical execution observations. Independent computational re-execution of the UCT/PUCT trajectories additionally requires their implementation, action-ordering conventions, prior settings, sampling and tie-breaking rules, random generator, seed, budget definition, software environment and input data. Those artifacts are not supplied with this article; thus full execution-level reproducibility is *not* claimed. That distinction should be explicit rather than hidden behind the word "reproducible." Furthermore, the trees are small and synthetic, the cited runs use seed 0, and no population-level error frequency follows. Some PUCT settings use informative prior weights, potentially oracle-derived. The certificate is retrospective and assumes closed, independently assignable unknown-leaf intervals and coherent treatment of shared variables.

Within those limits, two concrete propositions are well supported. First, four preserved selections can be compared against exact minimax values written on the page, exhibiting positive gaps. Second, five observed terminal values certify an ordering under incomplete information, while a specified deletion admits contrary completed worlds. A computational result is not automatically evidence for every conclusion attributed to it: *the conclusion depends on the information and the comparison standard actually supplied.*

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